TSM 099 Polynomials Test - Practice 1

Version 202610122020.19 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Factor completely over the integers, or write prime.

$(a)\quad 2x^2 - 5x - 3$

$(b)\quad 2x^2 + x + 3$

2. Multiply and simplify.

$(2x - 3)(3x + 2)$

3. Factor using the ac-method and grouping. Show ac, the split middle term, and the grouping.

$12x^2 + 29x + 15$

4. Factor completely.

$(a)\quad x^2 + 2x - 8$

$(b)\quad x^2 - 25$

5. Factor out the greatest common factor.

$4x^4 - 2x^3 + 6x^2$

Student answer sheet

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Question 1

(a) 10 pts

(b) 10 pts

Question 2

Polynomial 20 pts

Question 3

ac; pair 4 pts

Split middle term 4 pts

Grouping 6 pts

Factored form 6 pts

Question 4

(a) 10 pts

(b) 10 pts

Question 5

Factored form 20 pts

Answer key

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Question 1

(a)	$(x-3)(2x+1)$	10 pts
(b)	Prime	10 pts

Question 2

Polynomial	$6x^2-5x-6$	20 pts

Question 3

ac; pair	$ac=180;\ 9,\ 20$	4 pts
Split middle term	$12x^2+9x+20x+15$	4 pts
Grouping	$3x(4x+3)+5(4x+3)$	6 pts
Factored form	$(3x+5)(4x+3)$	6 pts

Question 4

(a)	$(x-2)(x+4)$	10 pts
(b)	$(x-5)(x+5)$	10 pts

Question 5

Factored form	$2x^2(2x^2-x+3)$	20 pts